Grokking as the Transition from Lazy to Rich Training Dynamics

Exercise Session - 1h, pen and paper

Reference: Kumar, Bordelon, Gershman & Pehlevan, ICLR 2024

Problem 1 - The Model and its Summary Statistics (10 min)

Consider a two-layer committee machine with N hidden neurons, input $\boldsymbol{x} \in \mathbb{R}^D$, and fixed readout weights all equal to 1:

$$f(\boldsymbol{w}, \boldsymbol{x}) = \frac{\alpha}{N} \sum_{i=1}^{N} \varphi(w_i \cdot \boldsymbol{x}), \qquad \varphi(h) = h + \frac{\epsilon}{2} h^2$$

The target function is $y(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x})^2$, with $\boldsymbol{x} \sim \mathcal{N}(0, \frac{1}{D}\boldsymbol{I})$.

(a) Define the two summary statistics:

$$\bar{\boldsymbol{w}} = \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{w}_i, \qquad \boldsymbol{M} = \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{w}_i \boldsymbol{w}_i^\top$$

Show that the network output can be written as:

$$f(\boldsymbol{x}) = \alpha\, \bar{\boldsymbol{w}} \cdot \boldsymbol{x} \; + \; \frac{\alpha\epsilon}{2}\, \boldsymbol{x}^\top \boldsymbol{M}\, \boldsymbol{x}$$

(b) What are the values of $\bar{\boldsymbol{w}}$ and $\boldsymbol{M}$ at random initialisation in the large N limit?

(c) For the network to perfectly fit the target on the test distribution, what must $\boldsymbol{M}$ and $\bar{\boldsymbol{w}}$ equal? (*Hint*: match the quadratic and linear parts of f to y.)

Problem 2 - The NTK of the Toy Model (10 min)

(a) Compute the NTK $K(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i=1}^{N} \nabla_{w_i} f(\boldsymbol{x}) \cdot \nabla_{w_i} f(\boldsymbol{x}')$ for this model. Show that, up to the overall prefactor α^2/N coming from this parametrisation, it can be expressed in terms of $\bar{\boldsymbol{w}}$ and $\boldsymbol{M}$ as:

$$K(\boldsymbol{x}, \boldsymbol{x}') \propto (\boldsymbol{x} \cdot \boldsymbol{x}') + \epsilon\, (\boldsymbol{x} \cdot \boldsymbol{x}')\, \bar{\boldsymbol{w}} \cdot (\boldsymbol{x} + \boldsymbol{x}') + \epsilon^2\, (\boldsymbol{x} \cdot \boldsymbol{x}')\, \boldsymbol{x}^\top \boldsymbol{M}\, \boldsymbol{x}'$$

(b) At initialisation ($\bar{\boldsymbol{w}} = 0$, $\boldsymbol{M} = \boldsymbol{I}$), the kernel simplifies to

$$K_0(\boldsymbol{x}, \boldsymbol{x}') \propto (\boldsymbol{x} \cdot \boldsymbol{x}') + \epsilon^2 (\boldsymbol{x} \cdot \boldsymbol{x}')^2.$$

This kernel is a sum of two terms. The first term $(\boldsymbol{x} \cdot \boldsymbol{x}')$ is sensitive to *linear* structure in the data, and the second $\epsilon^2(\boldsymbol{x} \cdot \boldsymbol{x}')^2$ to *quadratic* structure. Compute the ratio of the typical magnitude of the quadratic term to the linear term when x, x' are independent draws from $\mathcal{N}(0, \frac{1}{D}I)$. What does this tell you about how well-suited the initial kernel is for learning the (purely quadratic) target, especially when ϵ is small?

(*Hint*: use $\mathbb{E}[(\boldsymbol{x} \cdot \boldsymbol{x}')^2]$ for typical magnitude)

(c) The target $y(x) = \frac{1}{2}(\beta_\star \cdot x)^2$ is purely quadratic. After feature learning, the network can align $\boldsymbol{M}$ with $\beta_\star \beta_\star^\top$, effectively reducing the problem to learning a single direction in $\mathbb{R}^D$. Argue intuitively that this requires $P \sim D$ samples. By contrast, a fixed kernel that treats all quadratic directions equally would need far more samples. Why does this create an opportunity for grokking?

Problem 3 - Loss Decomposition (15 min)

The test MSE of the model is $\mathcal{L} = \langle (y - f)^2 \rangle$ where $\langle \cdot \rangle$ is the expectation over $x \sim \mathcal{N}(0, \frac{1}{D}I)$.

Useful identity (Isserlis' theorem for Gaussians). For $x \sim \mathcal{N}(0, \frac{1}{D}I)$ and any symmetric matrices A, B:

$$\langle (x^\top A\, x)(x^\top B\, x) \rangle = \frac{1}{D^2}\left[(\operatorname{Tr} A)(\operatorname{Tr} B) + 2\operatorname{Tr}(AB)\right]$$

In particular, $\langle (x^\top A\, x)^2 \rangle = \frac{1}{D^2}[(\operatorname{Tr} A)^2 + 2|A|_F^2]$ where $|A|_F^2 = \operatorname{Tr}(A^\top A)$.

(a) Decompose the network output as $f = f_{\text{lin}} + f_{\text{quad}}$ with $f_{\text{lin}} = \alpha \bar{\boldsymbol{w}} \cdot \boldsymbol{x}$ and $f_{\text{quad}} = \frac{\alpha\epsilon}{2}\boldsymbol{x}^\top \boldsymbol{M}\, \boldsymbol{x}$. Note that the target $y = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{\beta}_\star \boldsymbol{\beta}_\star^\top \boldsymbol{x}$ is purely quadratic. Use the fact that odd moments of isotropic Gaussians vanish (i.e. $\langle x_i\, g(x^\top A x) \rangle = 0$) to argue that the MSE splits as:

$$\mathcal{L} = \langle (y - f_{\text{quad}})^2 \rangle + \langle f_{\text{lin}}^2 \rangle$$

Then compute $\langle f_{\text{lin}}^2 \rangle$ explicitly. This is **Term C** (linear power).

(b) Now focus on the quadratic part. Define $A = \boldsymbol{\beta}_\star \boldsymbol{\beta}_\star^\top - \alpha\epsilon\, \boldsymbol{M}$, so that $y - f_{\text{quad}} = \frac{1}{2}\boldsymbol{x}^\top A\, \boldsymbol{x}$. Apply the Isserlis identity above to compute $\langle (y - f_{\text{quad}})^2 \rangle = \frac{1}{4}\langle (\boldsymbol{x}^\top A\, \boldsymbol{x})^2 \rangle$. Show that the result can be written as the sum of two terms:

$$\underbrace{\frac{1}{4D^2}(\operatorname{Tr} A)^2}_{\text{Term A: variance error}} + \underbrace{\frac{1}{2D^2}|A|_F^2}_{\text{Term B: misalignment error}}$$

Write out Term A and Term B explicitly in terms of $|\boldsymbol{\beta}_\star|^2$, $\operatorname{Tr}\boldsymbol{M}$, α, ϵ, and $|\alpha\epsilon\boldsymbol{M} - \boldsymbol{\beta}_\star\boldsymbol{\beta}_\star^\top|_F$.

(c) Interpret each of the three terms (A, B, C) in one sentence.

(d) At initialisation ($\bar{\boldsymbol{w}} = 0$, $\boldsymbol{M} = \boldsymbol{I}$), which terms are large and which are small? What happens in early training: recalling from Problem 2(b) that the initial kernel is much more sensitive to linear structure than quadratic structure, which component of f does the network use first to reduce train loss, and what does this do to the three terms?

Problem 4 — The Role of α: Controlling Laziness (10 min)

Recall the model $f(\boldsymbol{w}, \boldsymbol{x}) = \frac{\alpha}{N}\sum_{i=1}^{N}\varphi(\boldsymbol{w}_i \cdot \boldsymbol{x})$ with target $y(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x})^2$. The parameter α multiplies the network output but does **not** appear in the target. We train with gradient flow on the MSE loss $\mathcal{L} = \frac{1}{2}\sum_\mu (y_\mu - f_\mu)^2$.

Important note. In the paper, large-α experiments are implemented using the centered predictor

$$\tilde{f}(\boldsymbol{w}, \boldsymbol{x}) = f(\boldsymbol{w}, \boldsymbol{x}) - f(\boldsymbol{w}_0, \boldsymbol{x})$$

and the learning rate is rescaled as $\eta \propto \alpha^{-2}$. Here we give the basic scaling intuition in raw gradient-flow time.

(a) Scaling of the NTK. Compute the Jacobian $\nabla_{\boldsymbol{w}_i} f$ and show that it scales as α. Deduce that the NTK Gram matrix $(K_0)_{\mu\nu} = \sum_i \nabla_{\boldsymbol{w}_i} f(\boldsymbol{x}_\mu) \cdot \nabla_{\boldsymbol{w}_i} f(\boldsymbol{x}_\nu)$ scales as α^2.

(b) Kernel regression timescale. In the linearized (lazy) regime, the prediction dynamics are $\dot{f}_\mu = -\sum_\nu K_{\mu\nu}(f_\nu - y_\nu)$. Argue that the timescale for the training predictions to converge to the targets is:

$$t_{\text{kernel}} \sim \frac{1}{\alpha^2}$$

Note that the targets $y_\mu = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x}_\mu)^2$ are $O(1)$ — they are independent of α.

(c) Parameter displacement. The linearized model is $f_\mu \approx f_\mu^0 + \sum_i \nabla_{\boldsymbol{w}_i} f|_{\boldsymbol{w}_0} \cdot \delta\boldsymbol{w}_i$, where $\delta\boldsymbol{w}_i = \boldsymbol{w}_i - \boldsymbol{w}_i(0)$. To fit the training data, the network needs $\delta f_\mu \sim O(1)$. Using the fact that $\nabla_{\boldsymbol{w}_i} f \sim \alpha$, show that the required parameter displacement satisfies:

$$|\delta\boldsymbol{w}| \sim \frac{1}{\alpha}$$

Explain why this means large α keeps the network in the lazy regime.

(d) Feature learning timescale. The NTK depends on the parameters through $\varphi'(\boldsymbol{w}_i \cdot \boldsymbol{x})$. Show that the fractional change in the NTK during kernel regression is:

$$\frac{\Delta K}{K_0} \sim \frac{|\delta \boldsymbol{w}|}{|\boldsymbol{w}_0|} \sim \frac{1}{\alpha}$$

(using $|\boldsymbol{w}_0| \sim O(1)$ at standard initialization). Deduce that for large α, the NTK has barely rotated by the time train loss reaches zero: memorization happens **before** any significant feature learning.

For the NTK to rotate by $O(1)$ (which is what generalization requires), the parameters must change by $O(1)$. This happens on a much longer timescale than the initial kernel fit. A standard scaling summary is:

$$t_{\text{feature}} \sim \alpha^2$$

(e) The grokking gap. Compute the ratio $t_{\text{feature}}/t_{\text{kernel}}$ as a function of α. Describe what happens in the two limits $\alpha \to 0$ and $\alpha \to \infty$.

Problem 5 - The Role of ϵ: Initial Kernel Alignment (5 min)

Large ϵ means the initial kernel already places significant power on quadratic functions. Explain why this *reduces* or *eliminates* grokking. Conversely, why does small ϵ *increase* the grokking delay but lead to *lower* final test loss?

Problem 6 - Synthesis: Why Grokking is a Lazy-to-Rich Transition (5 min)

(a) State the three conditions identified by Kumar et al. that are jointly sufficient for grokking.

(b) Using the loss decomposition from Problem 3, describe the two-phase dynamics during grokking: - Phase 1 (early training): What happens to Terms A, B, C? - Phase 2 (late training): What happens to Terms A, B, C?

(c) Explain in 2-3 sentences why weight decay can help produce grokking but is not necessary, and how the lazy-to-rich framework subsumes weight-decay-based explanations.